Jupiter's moons as a measuring stick: both sides of Earth's orbit

What the geometry gives you, what it does not, and how Rømer timed the same stretch.

Faria · faria.jbf.com · jbf@jbf.com · 11 November 2026

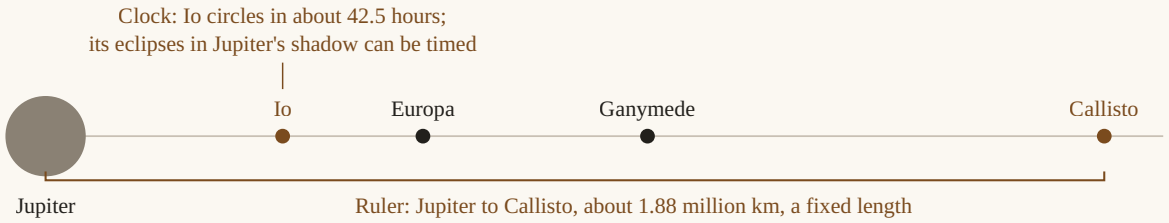
1. The moons are a clock and a ruler

Jupiter's four big moons keep time, and they keep a fixed size.

Galileo found them in January 1610. Outward from Jupiter they are Io, Europa, Ganymede, and Callisto.

The clock. Io goes around Jupiter in about 42.5 hours. When it enters Jupiter's shadow, the sunlight it reflects is cut off. That eclipse can be timed. Galileo proposed it as a longitude clock; that story is the lesson Galileo's moons.

The ruler. Each orbit keeps the same true size. Callisto, the outer moon, stays about 1.88 million km from Jupiter. When it swings farthest to one side (its *greatest elongation*), the gap from Jupiter to Callisto is a stick of fixed length.



Distances from Jupiter are to scale, and so is Jupiter. The moon dots are enlarged so they can be seen.

Figure 1. The four moons in a line, as we see them from Earth. Distances are to scale; dots are enlarged.

From Earth the moons' orbits are seen almost edge-on, so the moons look like a line of dots beside Jupiter. The ruler is the separation of one moon from Jupiter, not the width of its whole orbit.

2. Earth looks from two sides of its orbit

Over a year, Earth's distance to Jupiter changes by about the width of Earth's orbit.

Earth circles the Sun once a year on a slight ellipse. Jupiter is about 5.2 AU from the Sun. (1 AU is Earth's average distance from the Sun, about 93 million miles.)

- **Near side (opposition).** Sun, Earth, and Jupiter are nearly in a line; Jupiter is opposite the Sun in our night sky. Earth is about 5.2 − 1 = **4.2 AU** from Jupiter.
- **Far side (conjunction).** Jupiter is beyond the Sun. Earth is about 5.2 + 1 = **6.2 AU** away.

The difference, about 2 AU, is the width of Earth's orbit.

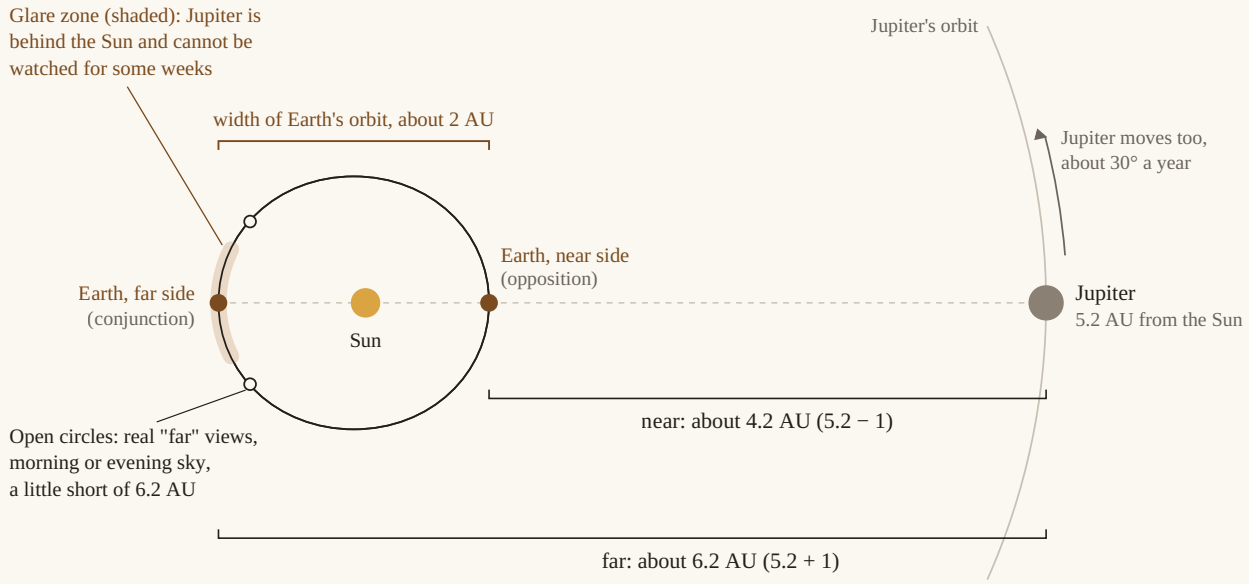


Figure 2. Seen from above. The far-side position is in the Sun's glare, so the real "far" views come a little before or after it.

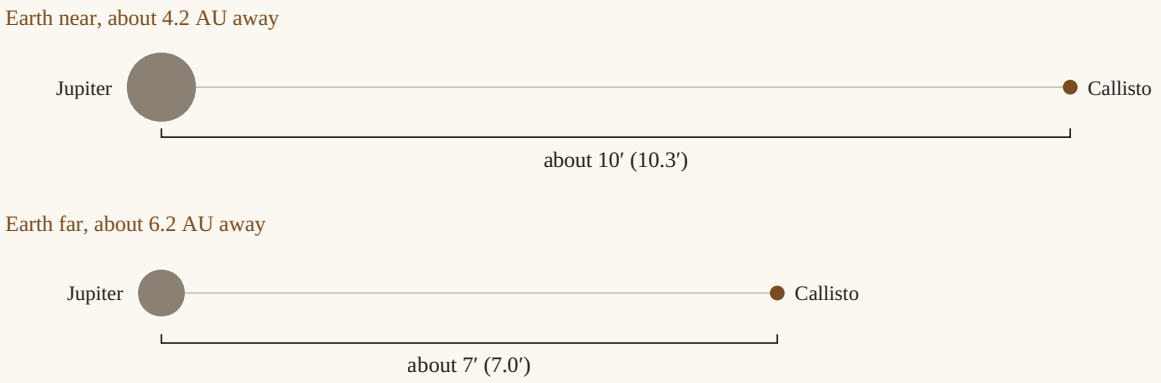
The catch. At the far side Jupiter is behind the Sun and lost in its glare for some weeks. No one can watch the moons there. Useful "far" views are taken when Jupiter is still in the morning or evening sky, a little short of 6.2 AU.

3. A fixed stick looks smaller when farther away

The angle you see is the stick's true length divided by its distance.

One handy fact: an object looks 1 arcminute (1′) wide when it is 1/3,438 of its distance. (A dime, seen face-on, at about 200 feet.) So:

> **Near.** 4.2 AU ≈ 628 million km. At that distance 1′ covers 628,000,000 ÷ 3,438 ≈ 183,000 km. Callisto's 1,880,000 km ÷ 183,000 ≈ **10′**.
>
> **Far.** 6.2 AU ≈ 928 million km. Here 1′ covers ≈ 270,000 km. 1,880,000 ÷ 270,000 ≈ **7′**.



The same 1.88-million-km stick at its greatest elongation, drawn at one angular scale. Jupiter's disc is to the same scale. Rounded, idealized numbers: Jupiter at its average 5.2 AU, Earth at 1 AU.

Figure 3. Same stick, two distances. 10′ is a dime, face-on, at about 20 feet; 7′ is the same dime at about 29 feet.

4. What the angles give you, and what they do not

Angles give ratios. They do not give miles.

- **Ratio.** 10.3′ ÷ 7.0′ ≈ 1.48. So the far distance is about 1.48 times the near one (6.2 ÷ 4.2 ≈ 1.48).
- **Difference.** Far minus near is about the width of Earth's orbit, 2 AU.
- **Together.** Near ≈ 2 ÷ (1.48 − 1) ≈ 4.2 AU, which fits Kepler's scale (Jupiter 5.2 AU from the Sun). This is a teaching check, not a historical method; small errors in the angles swing it a lot.

Every answer comes out in AU. The ruler cannot supply miles either: we know Callisto's 1.88 million km today only because we know the AU. In the 1600s only the stick's angle could be measured.

To turn the scale into miles you need one length measured on the ground. That came from other work: Cassini and Richer's Mars parallax of 1672, then the transits of Venus in 1761 and 1769. Today 1 AU = 149,597,871 km, about 93 million miles.

5. Rømer timed the same stretch

Same near-and-far geometry, measured in time instead of angle.

In 1676 Ole Rømer argued that Io's eclipses run late when Earth is farther from Jupiter and early when it is nearer, because light needs time to cross the extra distance. The full story is the Rømer lesson.

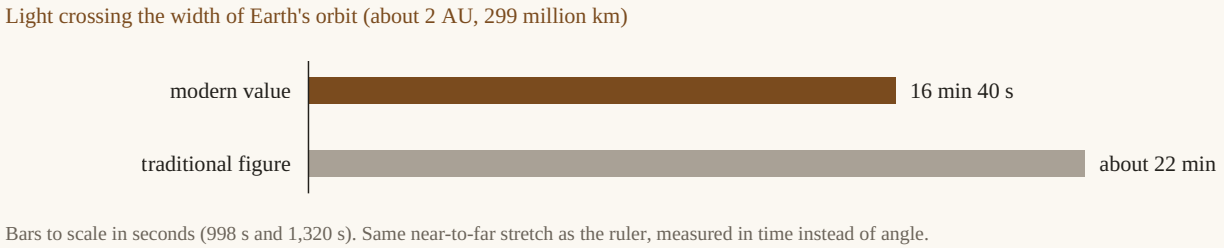


Figure 4. Modern: 299,195,742 km ÷ 299,792.458 km/s ≈ 998 s ≈ 16 min 40 s.

- The traditional figure is about 22 minutes to cross the diameter of Earth's orbit; the modern value is about 16 minutes 40 seconds.
- Rømer did not publish a speed. Huygens, in 1690, turned an orbital scale into a speed near 230,000 km/s. The defined speed today is 299,792.458 km/s.
- Once the AU is known in miles, those minutes become a speed. With a known speed, the same delay can be read as a distance.

Not two magic endpoints. You cannot take one near-side and one far-side observation and be done. Near conjunction the glare hides Jupiter, so the "far" data fall short of the full 2 AU. Jupiter moves about 30° a year. Earth's orbit is a slight ellipse (0.983 to 1.017 AU from the Sun). Io's orbit is not a perfect clock. The argument is a stack of dates. The 4.2 AU, 6.2 AU, 10′, and 7′ in this lesson are rounded, idealized teaching numbers.

Key facts

- Io is the clock (about 42.5 hours). Jupiter to Callisto, about 1.88 million km, is the ruler.
- Earth is about 4.2 AU from Jupiter on the near side and about 6.2 AU on the far side, a change of about 2 AU.
- Callisto's greatest elongation: about 10′ near, about 7′ far. Angle ≈ length ÷ distance.
- Angles give ratios in AU, not miles. Miles came from Mars in 1672 and the Venus transits of 1761 and 1769.
- Rømer (1676) timed the same stretch: traditional 22 min, modern 16 min 40 s. Huygens, not Rømer, gave a speed.

Sources

- Faria, "Galileo's moons: new centers and lasting names," faria.jbf.com/topics/64-galileo-jupiters-moons/
- Faria, "Rømer and Io: light needs time," faria.jbf.com/topics/65-romer-io-speed-of-light/
- Paris Observatory, "340 years ago, Romer showed that light has a finite speed," observatoiredeparis.psl.eu/340-years-ago-romer-showed.html
- Standard modern values: Jupiter 5.20 AU from the Sun; Callisto 1.883 million km from Jupiter; 1 AU = 149,597,871 km; c = 299,792.458 km/s.